# Behavior-Aware Multilingual Voice-Based Local Service Discovery and Workforce Empowerment System Using a Multi-Agent AI Framework

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Project Proposal Report

**Ihalagedara H T B – IT22274366**

B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

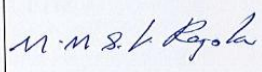
March 2026

## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

| Name | Student ID | Signature |
|---|---|---|
| Ihalagedara H T B | IT22274366 |  |

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

| Name | Date | Signature |
|---|---|---|
| Prof. Samantha Rajapaksa | 3/13/2026 |  |

# ABSTRACT

People now use online service platforms to access local services because these platforms have experienced rapid growth. Existing systems do not provide support for natural communication patterns and accessibility requirements which are essential in multilingual environments like Sri Lanka where users switch between Sinhala and English. The digital platforms of various local service providers such as plumbers and electricians and domestic workers do not exist because these workers lack technical skills and face difficulties in completing the registration process.

This research presents a local service discovery system which utilizes multilingual voice recognition together with behavior-driven intelligence and multi-agent artificial intelligence technology. The system enables service seekers and providers to interact through natural voice communication which supports multiple languages, while they handle urgent requests which enhance accessibility and reliability and inclusive access to services.

Multiple intelligent components exist within the solution. The multilingual voice intent model processes code-switching and incomplete speech while detecting urgency through acoustic signals which include pitch and speech rate and energy level variations. The system uses a cooperative multi-agent coordination layer which determines service provider locations and matches service demand needs and calculates their operational capacity and dependability and their urgent service requests. The AI-driven workforce intelligence module establishes fair job allocation methods while managing worker reputation and predicting employment needs to create sustainable job opportunities.

The system offers non-technical providers a voice-based onboarding system which requires no technical skills for use. Providers can create accounts and develop their profiles and use the platform by engaging in voice conversations which require no text writing or password usage. The framework supports multilingual voice prompts which use voice biometric authentication and OTP-based authentication methods through an HTML interface.

Keywords: Multilingual Voice Interaction, Multi-Agent System, Service Discovery, Behavior-Aware AI, Workforce Intelligence, Voice-Based Authentication, Digital Inclusion.

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# LIST OF ABBREVIATIONS

| | |
|---|---|
| **AI** | Artificial Intelligence |
| **ASR** | Automatic Speech Recognition |
| **CNN** | Convolutional Neural Network |
| **MFCC** | Mel-Frequency Cepstral Coefficients |
| **NLP** | Natural Language Processing |
| **NLU** | Natural Language Understanding |
| **SDG** | Sustainable Development Goal |
| **SLIIT** | Sri Lanka Institute of Information Technology |
| **TTS** | Text-to-Speech |
| **WER** | Word Error Rate |

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# 1. INTRODUCTION

## .1.1. Background and Context

The progress in Artificial Intelligence (AI), Natural Language Processing (NLP) and speech recognition technologies have enhanced human–computer interaction (HCI). Voice-based interface enables the user to interact with the digital systems through natural spoken language instead of the conventional written or graphical interfaces. Voice assistants have become very popular in the sphere of healthcare, transportation, and emergency services, and the increased significance of voice-driven intent recognition systems that are capable of correctly interpreting user requests and responding in real-time.

Nevertheless, multilingual systems of voice interaction continue to experience issues in the real world. In other nations like Sri Lanka, people often change languages in the same sentence, mixing both Sinhala and English in informal conversational language. Conventional voice recognition systems tend to be trained on monolingual and structured datasets, and therefore, are less effective with mixed languages commands or informal speech patterns. Therefore, there are numerous voice systems currently in place with the inability to effectively detect user intent in more than one language.

An additional difficulty arises in situations with a high level of stress, when users can give interrupted or partial speech instruction, like: “accident... hospital... fast... Traditional speech recognition systems are programmed to handle grammatically complete sentences, and thus, they cannot handle spontaneous and incomplete speech input. Proper intent identification and situational knowledge is thus highly crucial in developing intelligent systems that can be responsive to the needs of the user.

Moreover, most of the currently available voice assistants are mainly speech-to-text decoders and do not recognize any acoustic cues, including pitch shifts, speech rate, and voice energy. Such acoustic characteristics can be useful in telling us the emotional or situational status of the user, that of urgency or distress. The absence of systems that integrate such behavioral stimuli restricts the proper reaction of the existing voice systems to respond adequately in time-intensive scenarios.

The most recent research have discussed deep learning methods to understand user intent, and to such an extent that has included Bidirectional Long Short-Term Memory (BiLSTM) networks, which enhance intent prediction by considering the contextual relationship in speech or text sequences. Nevertheless, these strategies are still limited to catering to multilingual speech patterns, incomplete instructions, and urgency at the same time.

Thus, there is a gap in the studies of voice-based platform that can comprehend the code-switched multilingual speech, identify the urgency based on acoustic features, and discern the fragmented user instructions. Filling this gap is significant in applications where intent recognition can be accurate and fast to enhance accessibility of the services and efficiency in response.

In order to solve these problems, this study suggests an Adaptive Multilingual and Urgency-Aware Voice Intent Model that focuses on comprehending real-life spoken language patterns. The proposed model concentrates on three key capabilities; code-switch based multilingual intent recognition, fragmented speech recognition, and urgency recognition with the help of acoustic

cues like Mel-Frequency Cepstral Coefficients (MFCC), pitch, and speech energy. This component is the main voice understanding component of the system that retrieves user intent in verbal instructions and facilitates downstream service coordination functions.

This study correlates with the United Nations Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure since it advances innovative digital technologies and smart service systems. Also, it is related to SDG 3: Good Health and Well-Being, as it allows responding more quickly in the case of urgent or emergency situations.

Scholarly the proposed work belongs to the Machine Learning and Artificial Intelligence research cluster, in particular to the speech processing, natural language understanding, and intelligent human-centered AI systems. This study will make a contribution to the development of adaptive voice-driven services systems by addressing drawbacks in the currently available voice interaction technologies.

## 1.2 Literature Review

The recent evolution of artificial intelligence, natural language processing, and speech recognition has made it possible to create intelligent voice-based systems that may be used in more complex human-computer interaction situations. Innovations in modern service discovery typically combine conversational interaction models, machine learning models and contextual decision models to enhance the user experience and access to services. The studies in this area have been aimed at supporting systems to perceive the intent of the user via speech input, contextual information, and aligning suitable response via automated decision-making models. Nonetheless, even with the enormous advances, numerous current systems continue to be unable to process real-life conversational behavioral patterns, including multilingual speech patterns, incomplete commands, emotion, multi-modal inputs, and others. Such shortcomings lend credence to the necessity of more versatile and more situational voice-based systems that have the capacity to comprehend a variety of user communication patterns and enhance service discovery accuracy.

### 1.2.1 Code-Switching in Speech Recognition

Code-switching is a phenomenon of using two or more languages in an equal measure in a conversation or sentence. The latter is especially prevalent in multilingual communities, where languages naturally blend in everyday communication of its speakers. Conventional speech recognition systems are usually trained using monolingual data and therefore they are ineffective in multilingual inputs. Studies have indicated that multilingual speech recognition systems usually have lower accuracy rates in case of code-switched speech because of the difference in phonetics, grammar forms and vocabularies among different languages.

Deep learning techniques have been investigated to deal with this challenge. General neural network-based models like Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and transformer-based models have been used to enhance the performance of multilingual speech recognition. Such models seek to model contextual dependencies among

words and comprehend mixed language better. Most of such systems, however, continue to be trained using formal-language-based datasets, which restricts their capacity to address informal conversational patterns in speech which are usually used in the real-world like Sinhalese-English code-switching or Tamil-English code-switching.

Moreover, the currently available commercial voice assistants, including Google Assistant, Amazon Alexa, and Apple Siri, partially allow multilingual interaction, but cannot completely understand informal code-switched commands. In such systems, the users are usually obliged to change language settings or issue grammatically structured instructions. Consequently, there is still a necessity to have the localized models of speech recognition that can effectively process informal mixed-language instructions, which are widely spoken in multilingual societies.

### 1.2.2 IFR- Speech intent recognition is incomplete and fragmented.

In everyday context, particularly when conditions are emergent or of great stress, users often speak incomplete or in fragments as opposed to complete sentence. The conventional voice recognition systems are also based on structured commands; therefore they cannot recognize spontaneous or partial commands. Consequently, systems might not interpret user intent in the event that speech input is not grammatically complete.

The application of deep learning models to enhance intent recognition on incomplete speech has been investigated in the field of human-computer interaction studies. These sequence modeling frameworks have been shown to exhibit high success in identifying contextual correlation in speech sequences alongside predicting user motives more effectively and efficiently e.g. Bidirectional Long Short-Term Memory (Bist) networks. These models take input sequences in forward and reverse directions, and thus they are able to obtain significant contextual information with certain sections of the speech input removed.

In spite of these developments, most of the systems that are currently in use are still unable to capture user intent in situations where speech input is either too brief, ambiguous or half-spoken. Moreover, the majority of intent recognition systems do not process raw speech data but instead perform the analysis of textual representations of a transcription after speech. It is a drawback in cases where speech transcription can be partial or incorrect.

### 1.2.3 Urgency and Emotion Detection of Acoustic-Based Natures.

Interpretation of emotional and situational context of speech is also a significant part of smart voice interaction systems. Pitch variation, speech rate, intensity of energy, and tone are some of the acoustic characteristics that can offer important information on the emotional state of a user. To give an example, higher pitch and quicker speech rates can be taken as a sign of urgency or distress, whereas slower and calmer speech patterns can be viewed as a common conversation pattern.

Recent studies have investigated acoustic feature extraction methods including Mel-Frequency Cepstral Coefficients (MFCC) that can be used to analyze speech signals and recognize emotional states. The models of machine learning (like Support Vector Machines (SVM), Convolutional Neural Networks (CNN), and deep neural networks) have been employed to identify the speech emotions and find out the urgency level. With these strategies, the system will be able to perform more than just speech recognition since it will be able to utilize behavioral cues based on speech patterns.

Nevertheless, most of the voice assistants currently available are more oriented towards conversion of speech into text and lack dynamic analysis of acoustic features to identify urgency or emotional emotions. Consequently, existing systems might respond to urgent requests and normal requests equally, and this prejudice might restrict their efficiency in time-sensitive situations, including emergency support or urgent service orders.

### 1.2.4 Multimodal Service Classification.

The multimodal learning synthesizes data in various forms like speech, text, and images and enhances the accuracy of the decisions. On service-oriented websites, users can both describe problems verbally and submit visual evidence. As an illustration, when a user requests a repair service, he or she can narrate over the phone and provide an image of the damaged equipment.

Recent studies in multimodal artificial intelligence have shown that multimodal data can be highly beneficial to both classification and contextual decoding of information. Image feature extraction with deep learning models, including Convolutional Neural Network (CNN), is a common procedure, whereas speech and textual inputs are processed with recurrent or transformer-based models. The multimodal fusion methods which combine these representations can help the systems to classify the services and identify problems more accurately.

Although multimodal systems have some benefits, a significant number of current service discovery platforms are based on the use of a text or voice-based description only, and they do not include visual information. This drawback may result in the misclassification of the services especially in cases where the visual context is required to identify the precise nature of a problem.

### 1.2.5 Voice-Based Service Discovery Systems in the Existing Systems.

The use of voice-based service discovery platforms has gained usage in order to enhance ease and accessibility of service based applications. These systems enable the user to find services by voice command as opposed to navigation manually through application interfaces. Simple service discovery services are offered by popular voice assistants like Google Assistant, Siri, and Alexa which allow users to search for or book services or access information via spoken instructions.

A number of research prototypes have also been done where voice-driven service recommendation systems are presented that combine speech recognition with contextual information like, user location, service ratings and availability. Such systems tend to use machine learning models to correlate user demands and appropriate service vendors and give personal suggestions.

The current voice-based service discovery systems, however, can be characterized by simple matching a keyword or intent classification based on text after speech transcription. They hardly pay attention to the multilingual code-switching, interrupted speech input, and urgency recognition. Also, most systems do not have the mechanisms of integrating contextual cues like emotional signals or multimodal information in the identification of service requirements. Those constraints show the necessity of smarter voice-based systems that are able to comprehend advanced patterns of communication with users and give them a more adaptive service recommendation.

## 1.3 Research Gap

Even though a lot has been achieved in the field of speech recognition and voice-based human-computer interaction systems, there are still number of limitations of the approaches that are currently used in intent recognition. The current voice interaction systems are mostly modeled with the help of monolingual speech corpus and designed language patterns, which restrict their capacity to comprehend multilingual conversational language that is widely used in the real world. Users of a multilingual society like the Sri Lanka are often able to code-switch in their speech cycles: that is, multiple languages are used in one sentence. Nevertheless, existing models of speech recognition usually fail to recognize such mixed-language instructions correctly since they are mainly trained on modeled and grammatically organized data.

The other weakness of the current intent recognition systems is that it also requires complete user commands that are well structured. In the real world, especially when the user is under stress or when the circumstances are urgent, the user tends to provide fragmented or incomplete speech inputs. A traditional voice recognition system is never geared to interpret intent based on such partial commands and this may lead to incorrect intent recognition or slow response. Although there are deep learning models that have enhanced contextual meaning of speech sequences, a number of systems continue to rely on the use of text-based representations retrieved upon speech transcription and thus they are less capable of dealing with spontaneous conversational speech patterns effectively.

Additionally, the existing voice assistants have been mostly concerned with speech-to-text conversion and then text-based intent recognition and disregard the presence of acoustic cues in the initial speech signal. Vital elements of speech like pitch fluctuation, speech rate, and energy intensity may be considered as an effective indicator of the emotional state or urgency of a user. Nevertheless, these acoustic features are hardly incorporated into the intent recognition process with existing systems, and this restricts their capacity to recognize urgent situations and respond to the system in line with the situation.

These constraints demonstrate an apparent research gap in the implementation of adaptive voice intent recognition models that can be used to handle multilingual code-switching, disrupted speech input, and urgency detection simultaneously based on acoustic analysis. The existing solutions are still inclined to overcome these challenges separately and not using a new intent recognition system. Consequently, voice-based systems are still ineffective in the real-life conversation setting where there is a mixture of languages, half-complete commands, and prompts to urgency.

To fill this gap, this study reveals an Adaptive Multilingual and Urgency-Aware Voice Intent Model that combines multilingual speech comprehension, incomplete speech comprehension and acoustic-based urgency comprehension in the same model. The proposed model will enhance the robustness and generalizability of voice intent recognition systems by adding contextual speech processing and acoustic features analysis. This method brings a new twist with the combination of code-switch-aware language modeling, fragmented speech processing, as well as analysis of behavioral cues to allow better understanding of user intent in real-world communication with complexity.

# OBJECTIVES

## 2.1 Main Objective

To create and build an adaptive multilingual and urgency-aware voice intent recognition scheme that would be able to decipher code-switched speech, fragmented voice commands and urgency cues based on acoustic conditions in natural conversational settings.

## 2.2 Specific Objectives

To prepare a multilingual voice intent recognition system that could recognize language patterns of code-switched speech that are often used in conversations in Sri Lankan, especially Sinhala-English mixed languages.

To develop an intent prediction system that can understand fragmented and incomplete input speech allowing the system to extract the intent of the user even when the user only partially speaks a command or when the command is not fully grammatically constructed.

To isolate and process acoustic speech data, that is, Mel-Frequency Cepstral Coefficients (MFCC) that can be used to determine behavioral aspects on voice signals, that is, pitch change, speech rate, and energy amounts.

To create a urgency classification module to classify the request of users into urgency levels (e.g., normal, moderate, and high urgency) based on speech acoustic and contextual features.

In order to combine multilingual intent recognition and acoustic urgency detection into a single voice-understanding system, the system needs to be more accurate when it comes to interpreting real world spoken commands.

To test the performance of the suggested voice intent model in terms of intent classification, urgency detection and response processing time in simulated real-life voice interaction conditions.

# METHODOLOGY

## 3.1 Research Approach

In this study, an adaptive multilingual voice intent recognition model is designed and evaluated using both experimental evaluation and Design Science Research (DSR) approach. Design Science approach will be suitable since the research objective is to create a new technological artifact that can analyze the multilingual code-switched speech and identify the urgency based on the acoustic characteristics.

The research procedure consists of a number of steps: problem definition, model design, implementation, experimental analysis. First, the constraints of the current speech recognition systems in the fields of multilingual speech decoding, fragmented commands and urgency detection are analyzed. It is based on these gaps that the proposed voice intent model is implemented and designed.

Voice interaction datasets will be used to test the model after development and will represent real-world conversation situations. The results of the experiment will be compared with the baseline intent recognition techniques to determine the increase in intent recognition accuracy, urgency detection, and responsiveness of the system.

## 3.2 Technical procedures and algorithms.

The suggested element incorporates a variety of speech processing and machine learning methods that will allow proper voice intent identification.

### Speech Preprocessing

Preprocessing of voice input is done to enhance signal quality. Noise filtering, silence elimination, and audio normalization are examples of techniques used to decrease the background noise and pre-process the signal used in extracting the features.

Extraction of Acoustic Features.

The system uses Mel-Frequency Cepstral Coefficients (MFCC) to extract acoustic features of the speech, and it is the spectral properties of speech. Other characteristics such as pitch, speech rate and signal energy are also picked in order to obtain the behavioral cues connected with urgency.

### Multilingual Intent Recognition Model.

The system classifies intents with the aid of a Bidirectional Long Short-Term Memory (BiLSTM) neural network to understand multilingual and code-switched speech patterns. The speech sequences are fed to the BiLSTM models both forward and backward, with the forward direction processing speech sequences sequentially and presenting the context of words as well as enhancing intent detection accuracy.

The model is trained on annotated speech datasets with several intent types of service request, which involves a variety of scenarios.

Urgency Detection Module

The urgency classification model is built to find the level of urgency using acoustic features. Machine learning models like Rand Forest or the Support Vector Machines (SVM) are utilized to categorize the speech inputs as either an urgent or normal, moderate or high urgency.

### Intent–Urgency Fusion Layer

The results of the intent recognition model and urgency detection module are fused together in a decision fusion system. This will enable the system to generate a more context-sensitive meaning of user requests by combining the semantic intent and acoustic urgency indicators.

## 3.3 Tools and Platforms

Python will be used to implement the system, and it has comprehensive support of machine learning and speech processing libraries. The intent recognition model based on BiLSTM will be created with the help of deep learning frameworks, including TensorFlow or PyTorch.

The extraction of speech features will be done by means of Librosa and Open SMILE which support MFCC extraction and acoustic features analysis. The development of the models and their experimentation will be performed with the help of Jupiter Notebook environments, whereas the acceleration of the model training can be implemented with the help of Google Colab or the platforms powered by the GPUs.

These are also popular in speech processing studies, and they offer credible assistance in the development of experimental AI-based voice recognition systems.

## 3.4 Data Collection and Ethical Concerns.

The data set of this study will be made of multilingual speech samples comprised of Sinhala-English and Tamil-English code-switched commands. The speech data can be gathered by means of controlled recordings of the participants that are volunteers or be retrieved by publicly available speech datasets.

The participants will use full and partial speech patterns in providing voice commands regarding the service requests. All audio samples will be labeled with intent tags and urgency.

Each of the participants will give informed consent to the study. No personal identification data will be used and all recordings will be anonymized and kept securely to ensure the privacy of participants. The information will be utilized in the research only.

## 3.5 Strategy and Metrics of Performance of validation.

It is proposed that the proposed model will be tested by the means of an experimental validation approach. The data will be separated into training and testing subsets by use of train-test split or even k-fold cross-validation.

Accuracy, precision, recall and F1-score will be used to assess the performance of the intent recognition model. Classification performance with various intent categories will also be analyzed with the help of confusion matrices.

To detect urgency, classification accuracy and recall of high-urgency cases are going to be used in order to determine the capacity of the model to detect critical cases.

Also, the response of the system in time and the processing efficiency will be evaluated to determine whether the model will work with the real time voice interaction systems. The baseline speech recognition methods will be compared with the results to prove the efficiency of the proposed model.

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

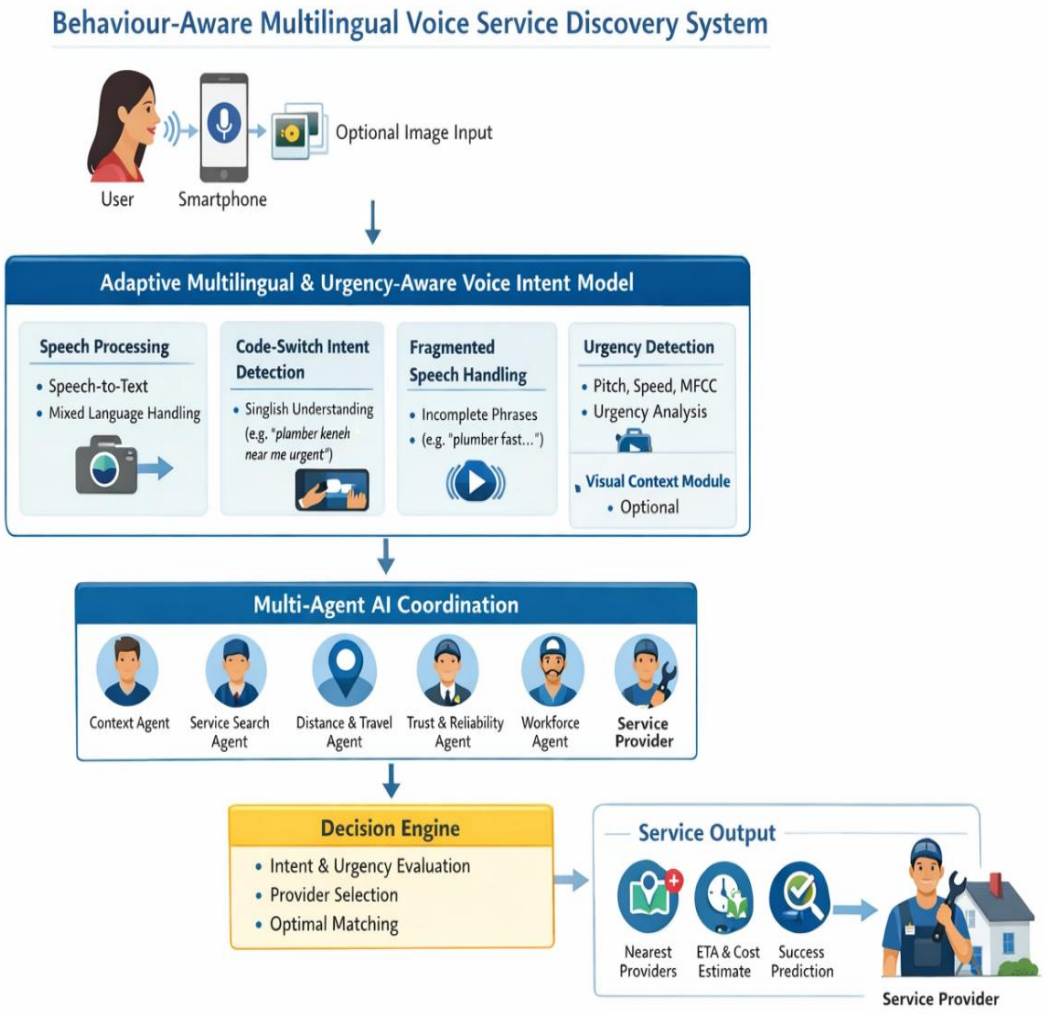



***Figure 1 - High level system architecture***

# 5.USER REQUIREMENTS

## 5.1 Stakeholder Identification

The identification of the stakeholders will be done in 5.1. The voice intent recognition component proposed entails a number of stakeholders, who interact or gain out of the system.

**Primary Users**

The major users are those that use voice commands to communicate with the system. Such users are allowed to order services in multilingual language by means of natural conversation patterns. The system will assist users who speak with the Sinhalese-English mixed speech, incomplete or urgent orders.

**System Developers / Researchers.**

The designers, trainers, and developers maintain, train and design the voice intent recognition model. Their interaction with the system occurs at the model development stage, training stage, testing stage, and performance evaluation stage. Service Platform Administrator: use of this category of administrator is limited to large organizations, and this position is not a common one.

**Service Platform Administrators:**

This type of administrator is only utilized in large organizations, and the position is not a typical role. The administrators control the entire service platform integrating this component of voice intent. Their activities include tracking the performance of the systems, keeping the bases of services and making sure that the platform works well.

**Service Providers (Indirect Stakeholders)**

Technicians or other professionals who will be involved in service delivery are indirect beneficiaries of the system since correct intent recognition will enhance service requests and minimal error communication between users and service providers.

## 5.2 Functional Requirements

The functional requirements indicate the special functions that the voice intent recognition component should undertake.

**FR1 – Voice Input Capture**

The system will record voice commands by the user via a voice interface or a microphone.

**FR2 – Speech Preprocessing**

The system will preprocess capture audio which includes noise reduction, silence elimination and normalize audio to enhance the quality of speech signal.

**FR3 Multilingual Speech Recognition**.

The system will be subjected to multilingual voice inputs of Sinhala-English or Tamil-English code-switched speech.

**FR4 – Fragmented Speech Interpretation**

The system shall detect and interpret user intent from incomplete or fragmented speech commands.

**FR5 – Acoustic Feature Extraction**

The system extracts acoustic features such as MFCC, pitch variation, speech rate, and energy levels from the speech signal.

**FR6 – Intent Classification**

The system will identify the voice commands of the users, based on the defined intent categories, by the Bist-based intent recognition model.

**FR7 – Urgency Detection**

Urgent intensity of user requests in the system shall be determined through the acoustic sense of speech.

**FR8 – Intent and Urgency Fusion**

The system will interpolate the information on intent and urgency that is detected to generate a final context-based voice interpretation.

**FR9 – Output Generation**

The system will produce structured intent outputs that may be used by the lower system components to coordinate services or generate responses.

## 5.3 Non-Functional Requirements.

Non-functional requirements are attributes of quality of the system and not the functionality. These requirements characterize the manner in which the system performs as opposed to what the system does.

### • Performance (Real-Time Processing)

Voice inputs should be processed by the system and intent predictions made within reasonable response time to facilitate real time voice interaction. It is a non-functional requirement as it characterizes the speed with which the system is to execute the intent recognition task and not a characterization of a particular system operation.

### • Accuracy and Reliability

The intent recognition and urgency detection models are supposed to be highly predictive in order to guarantee an accurate interpretation of the user commands. This has been regarded as a non-functional requirement in that it has to do with the quality and reliability of the output of the model rather than the functionality. Security and Data Protection The approach offers built-in protection against viruses, viral software, and other attacker types, while also preventing data loss and damage incurred from hacking (Mozilla, 2009). The system should secure the voice data that has been recorded and ensure that the voice datasets stored are not accessed by unauthorized parties. The entire amount of atmosphere speech data should be locked in and anonymized. This is not a functional requirement in the sense that it deals with the policies of system security and data protection, and not the functional aspects.

## • Scalability

The system is expected to be able to handle large amounts of voice samples and scale up training models without causing a major decrease in performance. This is a non-functional requirement since it concerns the capacity of the system to grow with the increased data and the use of the system rather than a particular operation.

## • Usability

The system must also offer easy voice interaction, where people are able to exchange with the system in a natural manner and do not have to use structured instructions. It cannot be functional as it explains the level of interaction with the user and usability, not an operational ability.

## • Robustness

The system is required to have consistent performance despite the background noise, mixed languages, or half sentences with voice inputs. This is not functional since it is defined as system resilience and stability in different conditions and not a given system task.

# 06. COMMERCIALIZATION PLAN

## 6.1 Customer Persona and Target Market.

The proposed voice intent recognition component is aimed at organizations and platforms that are based on voice-based interaction systems to enhance user-friendliness and effectiveness of services. The major market comprises of service discovery platforms, smart assistants developers, customer support automation systems, and digital service market places which demand intelligent voice interfaces that have the ability to comprehend multilingual speech.
In the multilingual areas like Sri Lanka and other countries in South Asia, people tend to speak mixed language conversational speech. Consequently, systems which offer on demand services, emergency services, home services, and customer support application are possible potential target markets of this technology.
The target customer profile will be comprised of technology firms and providers of service platforms who want to improve their voice engagement functionalities. The secondary users are the users of mobile applications that desire that their interaction can be done by natural voice as opposed to manual input in cases where communication is needed at a fast rate.

## 6.2 Value Proposition

The suggested system has a behavior conscious multilingual voice intent recognition model which enhances the accuracy and usability of voice-based service platforms. In contrast to the traditional voice assistants which are based on the structured command, the model under consideration will be able to decode code-switched multilingual speech, partial commands, and urgency signals based on the acoustic speech features.
This feature enables digital service platforms to respond more quickly and context-sensitive to user request, especially in emergency cases when the response to the request must be prompt. To increase the level of user experience, increase accessibility and operational efficiency of voice-based applications, the system increases understanding of voice commands and minimizes lack of understanding user intent.

## 6.3 Revenue Model

One strategy that the commercialization can use is the Software-as-a-Service (SaaS) model in which the voice intent recognition unit is offered as an API-based service to third-party platforms.

The usage of the model can be integrated into the mobile applications, service platforms, or voice-enabled systems of organizations in the form of a subscription-based API access model. Technology licensing agreements can also be used to generate revenue, and this is where firms pay to license the voice intent recognition technology to be used in their own systems. Moreover, tailor-made enterprise solutions may be provided to organizations that need to have a specific voice interaction system in the industries like healthcare, transport, or emergency services.

## 6.4 Cost Estimation

The main expenses that will be incurred in the development and implementation of the proposed system are the research and development expenses, computing infrastructure and the maintenance of the system. Costs of development are dataset preparation, training of machine learning model, and software implementation. Training and deployment of deep learning models can entail the use of computational resources like a GPU-enabled server or cloud solution.
Other expenses might be data collection, integration of system and/or optimization of model. The use of cloud-based infrastructure can be encountered by hosting the voice recognition API to use it in large scale. But after the model is created, the operations can be reduced by using scalable deployment of the clouds.

## 6.5 Pricing Strategy

The tiered subscription plans can be used as the pricing strategy of the proposed technology based on the level of system use. Small application developers can use the service on a simple subscription package with restricted API calls, and enterprise platforms may subscribe to more advanced packages with a more substantial processing capacity and customization.
The other possible pricing strategy is the usage-pricing strategy in which organizations pay in respect of voice requests made to the system. Cost flexibility is offered as the usage can be scaled according to the needs of the business without compromising the scale through this model.

## 6.6 Competitive Advantage

The solutions. The voice assistants used in most businesses only specialize in speech-to-text speech recognition and basic intent recognition and may not be able to handle multilingual speech and non-formal speech patterns.
The suggested model will provide a competitive edge since it will combine multilingual code-switching recognition, fragmented speech comprehension, and acoustic-based urgency detection into one framework. This enables the system to work better in real life conversational situations where the users can talk casually or can be in a hurry state.

The emphasis on voice understanding that is behavior aware also gives enhanced context awareness, and this allows smarter and more responsive voice-based service systems. proposed system has a number of benefits over the current voice recognition.

## 6.7. Intellectual Property considerations.

The proposed voice intent recognition model can consider new algorithms, model structure, and system integration methodologies that might be safeguarded using intellectual property systems. The strategies that can be used to protect against this are software copyright, patent applications on a distinctive model architecture or system workflow, proprietary datasets to train the model. Organizations that adopt this technology can also safeguard the software by licensing and limited API access that make the technology commercially controlled. The suggested system will be long-term commercialized and scaled with the help of proper intellectual property protection.

# 7. BUDGET AND JUSTIFICATION

The Adaptive Multilingual and Urgency-Aware Voice Intent Model is based on the use of resources in the collection of speech data, training of the machine learning model, cloud computing resources, and the preparation of data sets. The budget proposed concentrates on the key resources needed in the execution of the voice recognition model, undertaking acoustic feature analysis, and experimental evaluation. This component has been allocated an estimated amount of LKR 45,000, which is acceptable research cost range.

## 7.1 Hardware Costs

| Item | Description | Estimated Cost (LKR) |
|---|---|---|
| **External Microphone** | High-quality USB microphone for recording multilingual speech samples with minimal noise | 8,000 |
| **Headset with Noise Cancellation** | Used during testing and validation of speech recordings | 4,500 |
| **External Storage (Portable HDD / SSD)** | Storage for speech datasets, model checkpoints, and experimental results | 7,000 |
| **Total Hardware Cost** | | **19,500** |

***Table 01- Hardware Cost***

**Justification**

There must be hardware equipment to record the quality voice samples used to train and evaluate the speech recognition model. An excellent microphone should guarantee proper recordings of speech signals used to extract acoustic features of MFCC, pitch, and speech energy. Large audio datasets and trained machine learning models produced during experiments have to be stored in external storage.

## 7.2 Software Licenses

| Item | Description | Estimated Cost (LKR) |
|---|---|---|
| **Audio Annotation Tool (Optional License)** | Software used to label speech samples and annotate intent categories | 3,000 |
| **Speech Analysis Software Utilities** | Additional audio processing tools for analyzing acoustic signals | 2,500 |
| **Total Software Cost** | | **5,500** |

***Table 02- Software Licenses***

**Justification**

Even though numerous machine learning models, including Python, TensorFlow, PyTorch, and Librosa are open-source, structured speech annotation and audio analysis tools might entail small licensing fees. The tools are useful in formulating high quality labeled data needed to train intent recognition and urgency detection models.

## 7.3 Cloud Services

| Item | Description | Estimated Cost (LKR) |
|---|---|---|
| **GPU Cloud Computing (Google Colab Pro / Cloud GPU Credits)** | Required for training deep learning models such as BiLSTM networks | 10,000 |
| **Cloud Storage** | Temporary storage for datasets and trained models | 2,000 |
| **Total Cloud Cost** | | **12,000** |

***Table 03- Cloud Service***

**Justification**

Deep learning models used in speech processing are very computationally intensive and thus require a lot of computing power. Cloud GPU systems support quicker model training and allow large-scale its application to be carried out effectively. Datasets and outputs of experiments are also needed to be maintained in cloud storage throughout the development process.

## 7.4 Human Effort Cost

| Item | Description | Estimated Cost (LKR) |
|---|---|---|
| **Speech Data Annotation Assistants** | Assistance for labeling multilingual speech samples with intent categories and urgency levels | 5,000 |
| **Volunteer Compensation for Voice Recording** | Small compensation for participants providing multilingual speech recordings | 3,000 |
| **Total Human Effort Cost** | | **8,000** |

***Table 04- Human Effort Cost***

**Justification**

Machine learning algorithms need the correct training sets (that is, well-labeled). There is a need of human effort to label speech samples with the right intent tags and urgency. Furthermore, it might be necessary to ask the voluntary participants to voice-record multilingual voice commands to create a dataset. Small rewards will guarantee their regular attendance and quality records.

## 7.5 Total Budget Summary

| Category | Estimated Cost (LKR) |
|---|---|
| **Hardware** | 19,500 |
| **Software** | 5,500 |
| **Cloud Services** | 12,000 |
| **Human Effort** | 8,000 |
| **Total Estimated Budget** | **45,000 LKR** |

***Table 05- Total Budget Summary***

# 8. WORK BREAKDOWN STRUCTURE (WBS)

| ID | Task | Duration | Deliverable |
|---|---|---|---|
| **1.1** | Literature review and gap analysis | 2 weeks | Literature matrix |
| **1.2** | Corpus design and data collection planning | 1 week | Data collection protocol |
| **1.3** | Voice data collection (Sinhala-English, Tamil-English) | 3 weeks | Annotated speech corpus |
| **1.4** | ASR base model integration and code-switch fine-tuning | 4 weeks | Fine-tuned ASR model |
| **1.5** | Intent classification model development and testing | 3 weeks | Intent classifier v1 |
| **1.6** | Fragmented speech handling module development | 3 weeks | Incomplete speech model |
| **1.7** | Urgency classifier development and evaluation | 2 weeks | Urgency detection module |
| **1.8** | Visual context refinement module (optional) | 3 weeks | Multimodal fusion model |
| **1.9** | Pipeline integration and end-to-end testing | 2 weeks | Integrated voice pipeline |
| **1.10** | User evaluation (SUS testing) | 1 week | Evaluation report |
| **1.11** | Documentation and final report | 2 weeks | Final proposal/report |

***Table 06- Work Brekdown Structure***

# 9. GANTT CHART

The Gantt chart below shows the activities of Objective 1 as planned on a semester basis. The active period of development or evaluation of the tasks is depicted by each bar.

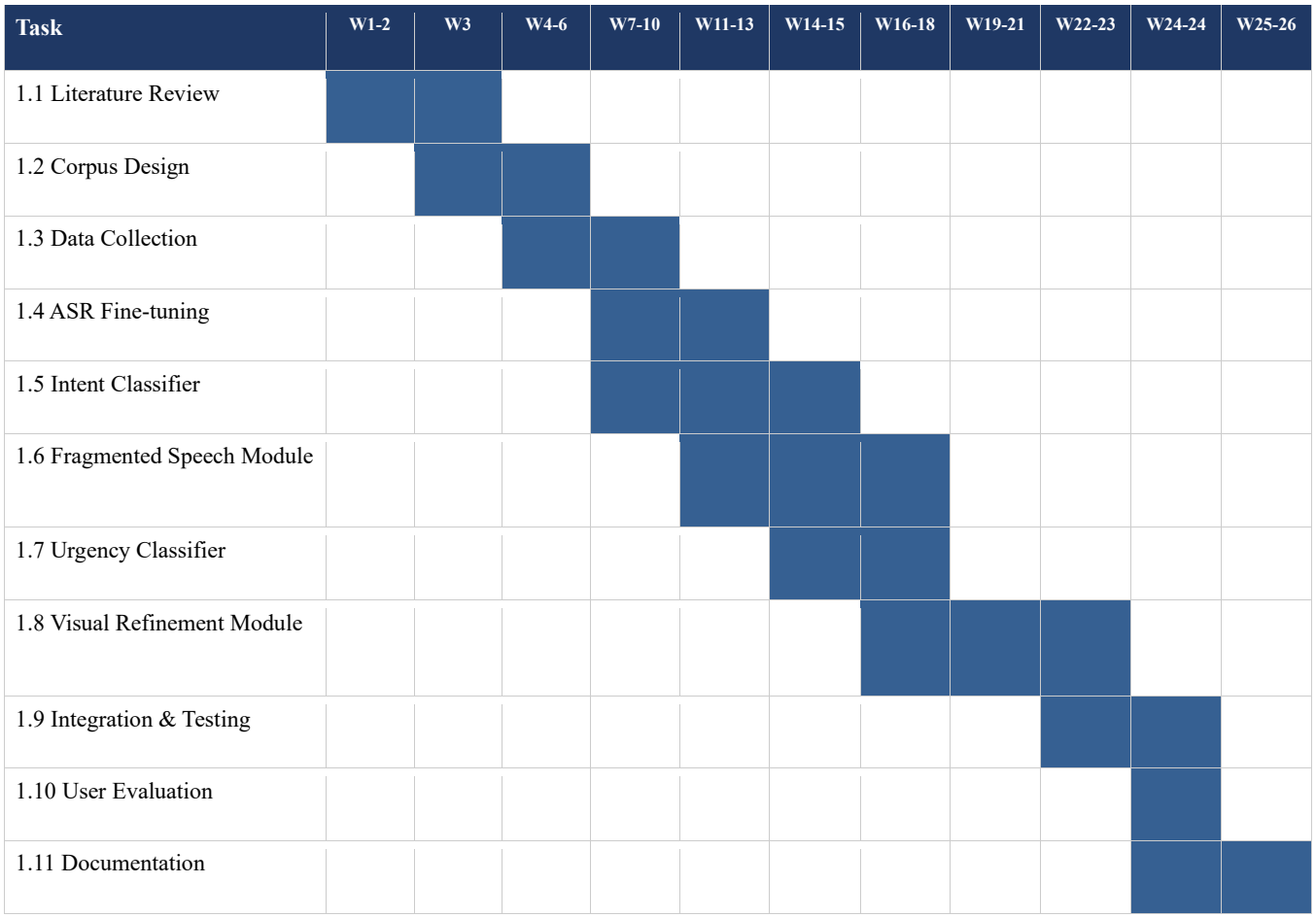



**Figure 2-** ***Gantt chart***

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# 11. APPENDICES

**Appendix A: Data Collection Consent Form Template.**

A written informed consent form will be given to the participants before all the voice recording sessions. The form will specify: (1) the aim of the research, (2) the character of the data gathered, (3) the way of how the data will be stored and utilized, (4) the right of the participant to withdraw any time as well as (5) the contact information of the research supervisor. Without signed consent, no recording is going to start.

**Appendix B: Sample Service Request Utterance Categories**

The following table shows examples of utterance categories that will be included in the training and test corpus:

| Category | Example Utterance | Urgency Level |
|---|---|---|
| **Medical (Emergency)** | *'Accident... hospital... fast...' (incomplete)* | High |
| **Medical (Normal)** | *'Hospital ekak near me one book karanna'* | Normal |
| **Plumbing** | *'Pipe leak tika fix karanna puluwanda?'* | Moderate |
| **Electrical** | *'Light switch work wen ne, electrician one'* | Normal |
| **Transport** | *'Taxi one now, urgent, airport yanawa'* | High |

***Table 07 – Appendix B***

**Appendix C: Risk Analysis**

| Risk | Likelihood | Impact | Mitigation Strategy |
|---|---|---|---|
| **Insufficient training data for Sinhala-English** | Medium | High | Augment with text-to-speech synthesis and transfer learning from similar language pairs |
| **Model latency exceeds 2s threshold** | Low | Medium | Apply model quantization and ONNX Runtime optimization |
| **Low volunteer participation rate** | Medium | Medium | Expand recruitment to multiple departments; offer course credit incentives |
| **GPU resource unavailability** | Low | High | Use free-tier cloud GPU (Kaggle/Colab); apply for academic GPU grants |
| **Ethical concerns around voice data** | Low | High | Strict consent protocols; immediate anonymization; SLIIT ethics board approval |

***Table 08- Appendix c***